

Towards Embodiment-Agnostic Human-to-Robot Motion Retargeting: A Position on Open Challenges

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Abstract

Enabling robots to learn from human demonstration is one of the most promising directions in robot learning, and motion retargeting (transferring human motion captured from wearable sensors or video onto a robot with a fundamentally different body) is a core component of it. Despite a decade of progress, most retargeting methods are built for a single robot platform and require substantial manual re-engineering to support a new one. This paper surveys the current landscape of human-to-robot motion retargeting and organises it into four families: geometric and kinematic mapping, optimisation-based retargeting, learning-based retargeting, and physics-aware imitation. From this survey we identify three recurring open problems (physics feasibility, morphology dependence, and input-quality sensitivity) and we argue for a shift toward embodiment-agnostic pipelines: systems that accept arbitrary human motion input and produce physically valid trajectories for arbitrary target robots without per-robot tuning. Recent methods have begun to close the first two of these gaps; we show that their integration with robust multi-modal input remains the open frontier. We close with a concrete research question and a plan for evaluating such a pipeline.

1. Introduction

Robots that can learn directly from human demonstration could remove much of the expert programming that currently limits their deployment in manufacturing, healthcare, and service settings. The underlying idea is simple: record what a person does using sensors or cameras, and transfer that motion to a robot. The difficulty lies entirely in the transfer. Human and robot bodies differ in their joints, link lengths, and degrees of freedom, so a direct copy of joint angles almost never produces a valid robot motion. Bridging this gap is the

problem of motion retargeting.

The field has advanced rapidly in recent years, but it has matured narrowly. Methods are typically demonstrated on one robot (a Pepper, a NAO, a single dexterous hand, or one full-sized humanoid) and the mapping, cost weights, or training data are tailored to that platform. Moving to a new robot usually means re-doing much of the engineering. We argue that this per-robot specialisation, rather than any single technical bottleneck, is now the principal obstacle to scaling demonstration-based robot learning.

This paper makes three main contributions. First, we provide a structured taxonomy of retargeting methods grounded in fifteen representative works spanning robotics and computer graphics, extended with recent cross-embodiment and physics-aware methods and summarised in a comparison table. Second, we identify three open problems that cut across all of them and assess how far recent work has come toward closing each one. Third, we take the position that the next advance must come from embodiment-agnostic retargeting, and we frame a concrete research question and evaluation plan to pursue it.

2. Background

2.1. What is retargeting?

Retargeting is a way of transferring motion from one source to another (in our case the source is a human and the target is a robot) while preserving the intent of the motion and respecting the robot’s constraints. The input may come from a wearable inertial suit, optical motion capture, or video-based pose estimation. The output is a sequence of robot joint-angle trajectories that should be both kinematically valid and physically executable.

2.2. The embodiment gap

The central obstacle is the *embodiment gap* between the human and the robot. It appears in three forms. There is a *structural mismatch*, because the human’s ball-and-socket joints rarely correspond cleanly to a robot’s revolute joints. There is a *degree-of-freedom mismatch*, because a robot segment may have fewer or more controllable axes than the human limb it imitates. And there is a *feasibility gap*, because a motion that is comfortable for a person may violate the robot’s joint limits, cause self-collision, or be dynamically impossible to execute.

2.3. Defining embodiment-agnostic retargeting

We use the term *embodiment-agnostic* to describe a retargeting pipeline that satisfies three properties. First, it accepts a robot’s kinematic and dynamic description (e.g. a URDF file) as runtime input rather than encoding robot-specific parameters during training or design. Second, it produces trajectories that are physically feasible for that robot without per-robot cost tuning or manual correspondence design. Third, it can be applied to robots of structurally different classes, such as a serial robot arm, a humanoid torso, or a dexterous hand, without architectural changes to the pipeline. This is deliberately a stronger requirement than multi-robot generalisation within a single structural class (e.g. humanoid-to-humanoid), which several existing methods already achieve.

3. Survey of Existing Approaches

3.1. Geometric and kinematic mapping

The most direct approaches define an explicit geometric correspondence between human and robot. Khalil et al.’s Pepper pipeline scales human limb vectors proportionally to the robot’s links, which is cheap and runs from ordinary video, but is sensitive to the noise and depth ambiguity of single-camera pose estimation, limiting it to expressive, low-precision motion [1]. Darvish et al. generalise this idea with whole-body geometric retargeting, which formulates the correspondence as inverse kinematics over the full robot model and is explicitly designed for scalability: different robots can be teleoperated by different users with only minimal changes to the system [3]. This scalability goal is precisely the direction this paper looks into, although the method still depends on a hand-designed correspondence per robot.

In real-time teleoperation, the same kinematic-

mapping idea is tuned for responsiveness and naturalness rather than offline data generation. Mohan and Kuchenbecker’s OCRA (Optimization-based Customizable Retargeting Algorithm) maps an operator’s upper-body motion from a commercial inertial suit onto a NAO humanoid in real time, deliberately exploiting the redundant degrees of freedom that end-effector-only methods discard, so that the robot’s whole arm, not just its hand, moves in a human-like way [2]. Such systems prioritise perceptual quality and are well suited to social robotics, but they are not intended to generate training data for autonomous policies.

3.2. Optimisation-based retargeting

Optimisation-based methods cast retargeting as a constrained problem: minimise the discrepancy between human and robot pose while satisfying joint limits and avoiding collisions. Wang et al. follow this template for a NAO humanoid, acquiring human motion from inertial measurement units and combining kinematic modelling, inverse kinematics, constraint enforcement, and a final difference-optimisation step to produce balanced, human-like upper-limb motion [4]. The approach achieves a good kinematic match but is computationally heavy and requires its cost terms to be tuned for the target robot.

The same paradigm scales to dexterous hands. DexPilot teleoperates a 23-degree-of-actuation hand-arm system from nothing more than four depth cameras observing a bare human hand: a nonlinear optimiser aligns a set of wrist-to-fingertip and inter-finger vectors between the human and robot hands via forward kinematics, producing fine manipulation behaviour without gloves or markers [5]. DexPilot shows both the power and the cost of optimisation-based retargeting; it is accurate and general within a hand morphology, but the vector correspondences and objective are made specifically for that hand.

3.3. Learning-based retargeting

Learning-based methods replace manual, hand-designed rules with mapping systems learned directly from data. The goal is to automatically teach the system what movements are physically possible for a robot, making it easier to adapt to new robot designs. The main difference among these methods is how much matched human-robot data they need to train.

Zhang et al. map human movements to robots by using a decoder network that searches a compressed latent space to find the best-matching robot pose. By modelling both humans and robots as con-

nected structural graphs, they successfully transferred complex, two-armed sign language to three different robots (YuMi, NAO, and Pepper) [6]. To address the problem of scarce training data, Choi et al. introduced a self-supervised method that automatically generates its own matching data. Crucially, they encoded mathematical boundaries in their data mapping to guarantee that every generated movement is within safe position limits and completely free of self-collisions [7].

Kim and Lee’s C-3PO system uses deep reinforcement learning in a continuous three-phase loop that refines movements with a human evaluator in the loop, demonstrating control of NAO, Pepper, and Baxter robots with a single framework [8]. Yan et al. created ImitationNet, which completely removes the need for paired data by building a shared space where human and robot movements are compared using a universal similarity metric, allowing the robot to be controlled by text, regular video, or simple poses [9]. The authors later extended this to dexterous hands using standard video [10]. These learning-based approaches complement foundational work on robot learning from demonstration, which has established the broader framework for encoding task-level behaviour from human examples [14, 15].

3.4. Physics-aware imitation and cross-morphology retargeting

This approach treats retargeting not just as a problem of matching body positions, but as the first step in physical control. It acknowledges that a mathematically perfect movement is useless if a physical robot cannot actually execute it without falling over or breaking.

DeepMimic is the foundational work in this area, using reinforcement learning to imitate reference motion clips and demonstrating that training a system with physics in mind produces stable, natural movements [11]. HumanMimic applies this concept to a real, full-sized humanoid robot (JAXON), pairing a simplified skeleton-mapping step with an adversarial scoring system that forces the robot to stay within safe, natural movement boundaries. It even allows the robot to naturally transition between gaits not explicitly shown in the training data [12].

The research that comes closest to handling any type of robot body originates in the computer graphics field. ACE (Adversarial Correspondence Embedding) retargets human movements onto characters with structurally different bodies, such as a four-legged creature with a robotic arm, a crab, and a wheeled robot with an extending arm [13]. ACE proves that a single framework can bridge completely different body

shapes, a capability that robotics still struggles with, but its main limitation is a focus on visual realism rather than the physical-contact constraints required by actual hardware.

3.5. Comparison of surveyed methods

Table 1 summarises the surveyed methods along five dimensions relevant to embodiment-agnostic retargeting.

The table makes the core gap visible: among the methods published before 2025, no approach simultaneously achieves multi-class generalisation, physics feasibility, and freedom from per-robot tuning. ACE comes closest on the first two criteria but is limited to animated characters; Choi et al. and ImitationNet eliminate per-robot tuning but stay within a single structural class and do not enforce physical feasibility. As the lower rows of the table show, recent methods have begun to close this gap, and we examine them next.

3.6. Recent progress toward embodiment-agnostic retargeting

A wave of 2025–2026 methods has begun to close the gaps our survey identifies. On the morphology-dependence front, several methods now retarget to many embodiments from a single representation. AdaMorph maps human motion into a morphology-agnostic latent space and decodes it with embodiment-aware adaptive normalisation, serving twelve humanoid platforms without per-robot training [19]. G-DRaM conditions a diffusion model on the target robot’s kinematic graph and generalises across heterogeneous skeletal structures [20]. Yan and Lee learn a unified latent space for cross-embodiment control, in which a new robot is added by learning only a lightweight embedding layer [21]. CrossDex applies the same spirit to dexterous hands, retargeting grasps across four hand morphologies via embodiment randomisation in reinforcement learning [22].

On the physics-feasibility front, retargeting is increasingly performed inside a physics simulator rather than as a purely geometric step. ReActor jointly adapts a reference motion to the target morphology while training a tracking policy with reinforcement learning, eliminating foot sliding, self-collisions, and dynamically infeasible motion from only sparse semantic correspondences [16]. X-Morph retargets human motion to non-humanoid legged platforms (a quadruped, a hexapod, and a quadruped manipulator) through a physics-aware corrector and a distilled tracking policy [18]. Spatio-temporal retargeting produces physically feasible, agile

Table 1. Comparison of surveyed human-to-robot motion retargeting methods. Multi-class = generalises across structurally different robot families. Phys. feas.: None = not addressed; Constr. = constraint/IK optimisation; Sim. = physics simulation or RL.

Method	Input	Target robot(s)	Multi-class?	Phys. feas.	Per-robot tuning?
Khalil et al. [1]	Video (monocular)	Pepper (humanoid)	No	None	Yes
Darvish et al. [3]	Wearable suit	Multiple humanoids	No	Constr.	Yes (per robot)
Mohan & Kuchenbecker [2]	Wearable suit	NAO (humanoid)	No	Constr.	Yes
Wang et al. [4]	IMU suit	NAO (humanoid)	No	Constr.	Yes
DexPilot [5]	Depth cameras	LEAP/Allegro hand-arm	No	Constr.	Yes
Zhang et al. [6]	Video / MoCap	YuMi, NAO, Pepper	No	None	Partial
Choi et al. [7]	MoCap	Humanoid (generic)	No	Constr.	No
C-3PO [8]	Video	NAO, Pepper, Baxter	No	None	Partial
ImitationNet [9]	Video / text	Humanoids (multiple)	No	None	No
Yan & Lee [10]	Video	Dexterous hands	No	None	No
DeepMimic [11]	MoCap	Simulated character	No	Sim.	Yes
HumanMimic [12]	MoCap	JAXON humanoid	No	Sim.	Yes
ACE [13]	MoCap	Animated cross-morph chars	Yes	Sim.	No
<i>Recent methods (2025–2026)</i>					
ReActor [16]	MoCap / video	Humanoids + quadruped	Yes	Sim.	No (sparse)
X-Morph [18]	Video / text	Quadruped, hexapod, quad-manip.	Yes	Sim.	Partial
AdaMorph [19]	MoCap	Humanoids (12)	No	Constr.	No
G-DReaM [20]	MoCap	Heterogeneous skeletons	Yes	None	No
Yan & Lee [21]	MoCap	Humanoids (multiple)	No	None	No
CrossDex [22]	MoCap	Dexterous hands (4)	No	Sim.	No
Yoon et al. [17]	Video	Quadrupeds (4)	No	Sim.	Yes

motion on four real quadruped robots of differing dimensions [17].

These methods substantially shrink the first two open problems below. What they do not yet do is combine them, and none of them addresses the robustness of the input side: each assumes clean, high-quality motion capture or video, and the work concentrates on humanoids and legged robots rather than on the serial manipulator arms and dexterous hands that a truly general pipeline must also cover. We formalise this remaining gap in the next section.

4. Open Problems

Three problems recur across these four families, and recent work has begun to close the first two while leaving the third largely untouched.

The first is **physics feasibility**. Most geometric, optimisation-based, and learning-based methods produce trajectories that are geometrically valid but dynamically impossible to execute. A trajectory that exceeds a robot’s joint torque limits or destabilises its balance is functionally useless. Only physics-aware methods like DeepMimic and HumanMimic confront this issue directly, and a new generation of methods (ReActor, X-Morph, and spatio-temporal retargeting for quadrupeds) now performs retargeting inside a physics simulator, largely eliminating foot sliding, self-collisions, and infeasible motion. These advances, however, target locomotion and whole-body tracking; the

arbitrary manipulation and hand morphologies that a general pipeline must also cover remain unaddressed.

The second is **morphology dependence**. The majority of surveyed methods are restricted to a single hardware platform, and those that generalise do so within one structural class, such as humanoids with similar skeletons or, in the case of ACE, animated characters. Recent unified models (AdaMorph, G-DReaM, and cross-embodiment latent-space control) now retarget to many embodiments from a single representation, and X-Morph extends this to non-humanoid legged platforms. Still, no method yet spans the full range from serial arms to humanoids to dexterous hands without some form of per-embodiment engineering, whether a robot-specific prompt, an embedding layer, or a correspondence set.

The third is **input-quality sensitivity**. Video-based approaches degrade rapidly under poor lighting or physical occlusion, while wearable motion-capture suits are robust to the environment but impose heavy calibration burdens. None of the recent methods above addresses this: they all assume clean, high-quality motion input rather than fusing video and wearable sensing robustly. No current pipeline integrates both input modalities, making this the clearest remaining gap.

5. Position: The Case for Embodiment-Agnostic Pipelines

Recent work has begun to deliver the abstraction we call for: ReActor and X-Morph make physics-feasible retargeting across different body classes achievable, and AdaMorph and G-DReaM show that a single model can serve many embodiments. Because single-robot tracking is largely solved, we argue that the remaining breakthrough lies not in any one component but in their integration: a unified pipeline that maps any human input to any target robot morphology, enforces physical feasibility, and tolerates the noisy, occluded, or sensor-based input that real deployment produces, without robot-specific tuning.

The missing link is a robot-independent human motion representation combined with a feasibility layer that automatically adapts to a target robot’s constraints. By robot-independent representation we mean a body-relative kinematic encoding, such as normalised joint directions, velocity fields, or contact events, that captures the intent of a motion without committing to any particular skeleton structure. This representation would be consumed by a feasibility layer that reads the target robot’s URDF description at runtime and applies the appropriate joint-limit and collision-avoidance constraints, rather than having them baked in at training time.

This leads to our research question: *Can a single retargeting pipeline generate physically feasible trajectories across multiple robot morphologies, from wearable-suit input, without per-robot re-engineering?* Recent methods answer parts of this question in isolation; none yet answers it in full, because none couples cross-morphology generality with robust multi-modal input.

We propose evaluating this by driving both a UR5 and a Franka Emika Panda arm in simulation using motion data from wearable suits. The recent cross-embodiment methods above have concentrated on humanoids and legged robots, leaving serial manipulator arms, the workhorse of manipulation, largely untested; this makes arms a meaningful first target. These two arms were selected as a deliberately controlled first test: despite both being serial manipulators, they differ meaningfully in their kinematic structure. The UR5 has a spherical wrist with six revolute joints and a long reach, while the Franka has a different joint-angle layout and a compliance-oriented design. This makes them a non-trivial pair within a well-defined class before scaling to more radically different platforms such as humanoids or dexterous hands. Feasibility will be enforced via the physics simulator and benchmarked

against a classical inverse-kinematics baseline. Success across these two arms with zero custom tuning would provide the first empirical evidence for a truly embodiment-agnostic pipeline; failure would reveal precisely where the representation or feasibility layer breaks down.

6. Conclusion and Future Work

This paper surveyed geometric, optimisation, learning, and physics-aware motion retargeting, identified physical feasibility, morphology dependence, and input sensitivity as core open challenges, and summarised key distinctions between methods in a structured comparison table. We further surveyed the recent cross-embodiment and physics-aware methods that have begun to close the first two of these gaps, and we showed that none yet combines them with robust multi-modal input. We defined embodiment-agnostic retargeting precisely: a pipeline that takes a robot’s kinematic description as runtime input, applies a robot-independent human motion representation, and enforces physical feasibility automatically, without architectural changes or per-robot tuning across structurally different robot classes.

Future work will develop and evaluate a pipeline of this kind using wearable-suit data to control simulated UR5 and Franka arms, benchmarked against classical inverse kinematics. Because recent work has focused on humanoids and legged robots, serial manipulator arms remain an open and under-explored target. A natural subsequent step would extend evaluation to more dissimilar platforms such as humanoids and dexterous hands, and investigate robust integration of video-based pose estimation as a complementary input modality.

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